

Symmetries: $P(R \rightarrow R_n) = |\langle \psi, \psi_n \rangle|^2$

• For different observers R and R'

$$P(R \rightarrow R_n) = P(R' \rightarrow R'_n)$$

• Unitary: $UU^\dagger = U^\dagger U = \mathbb{I}$

$$\langle U\psi, U\psi \rangle = \langle \psi, \psi \rangle \quad (\text{Unitary})$$

$$\langle U\psi, U\psi \rangle = \langle \psi, \psi \rangle^* \quad (\text{Anti-unitary})$$

or

$$U^\dagger = U^{-1}$$

• trivially: $U = \mathbb{I}$

• A rotation, translation or Lorentz can be made trivial (at each pt in time) and so $U \cong \mathbb{I}$

and so the above must be linear, unitary

• For a group element U

$$U: \text{End}(L^2) \cong G$$

we have a Lie alg. element T

s.t.

$$U = \mathbb{I} + i\epsilon T = e^{i\epsilon \tilde{T}}$$

* IF $U^\dagger U = \mathbb{I}$, then

$$(\mathbb{I} - i\epsilon T^\dagger)(\mathbb{I} + i\epsilon T) = \mathbb{I}$$

$$\mathbb{I} - i\epsilon T^\dagger + i\epsilon T = \mathbb{I}$$

$$T^\dagger = T \quad (\text{Self-adjoint / Hermitian})$$

$$* U(T) = e^{i\epsilon T}$$

$\hat{\hookrightarrow}$ reps of group elements

$$\rho: G \rightarrow \text{End}(V)$$

$$\rho(g) = U(T)$$

$$\text{where } g = e^{i\epsilon T}$$

• If $T_1: R_n \rightarrow R'_n$

$$R_n \equiv \{ \psi_n \}$$

$$U(T_1)\psi_n \in R'_n$$

$$R_n \xrightarrow{U_1} R'_n \xrightarrow{U_2} R''_n$$

$$U_1\psi_n$$

$$U_2 U_1 \psi_n$$

But note that

$U(T_2 T_1)\psi_n \in R''_n$ as well and can differ only by a phase $\phi_n(T_2, T_1)$

$$U(T_2 T_1) = e^{i\phi_n(T_2, T_1)} U(T_2) U(T_1)$$

- Let $\psi_A \in R_A$ $\psi_B \in R_B$ s.t.
 $(\psi_A, \psi_B) = 0$

$$\psi_{AB} = \psi_A + \psi_B$$

$$e^{i\phi_{AB}} U(T_2 T_1)(\psi_A + \psi_B) =$$

$$U(T_2) U(T_1) \psi_A + U(T_2) U(T_1) \psi_B$$

$$= e^{i\phi_A} U(T_2 T_1) \psi_A + e^{i\phi_B} U(T_2 T_1) \psi_B$$

Since U is unitary, $\exists U^{-1}$

$$e^{\pm i\phi_{AB}} (\psi_A + \psi_B) = e^{\pm i\phi_A} \psi_A + e^{\pm i\phi_B} \psi_B$$

By linear independence

$$e^{i\phi_{AB}} = e^{\pm i\phi_A} = e^{\pm i\phi_B}$$

thus

$$\phi_n(T_2, T_1)$$

is independent of which state vector we are looking at:

$$\phi_n(T_1, T_2) \rightarrow \phi(T_1, T_2)$$

$$U(T_2)U(T_1) = e^{i\phi(T_2, T_1)} U(T_2 T_1)$$

$\phi = 0$: U is a representation
 $\phi \neq 0$: projective representation

• Assume $\phi = 0$:

ex: Poincare'

$$x' = \Lambda_1 x + a_1$$

$$U(\Lambda_1, a_1)x = \Lambda_1 x + a_1$$

$$U(\Lambda_2, a_2)U(\Lambda_1, a_1)x$$

$$U(\Lambda_2, a_2)(\Lambda_1 x + a_1) \leftarrow x'$$

$$U(\Lambda_2, a_2)x' = \Lambda_2 x' + a_2$$

$$U_2 U_1 x = \Lambda_2 \Lambda_1 x + \Lambda_2 a_1 + a_2$$

$$U(\Lambda_2, a_2)U(\Lambda_1, a_1) = U(\Lambda_2 \Lambda_1, \Lambda_2 a_1 + a_2)$$

Poincare is a semi-direct product

$$[O^\uparrow(1,3) \ltimes_p \mathbb{R}^{1,3}]$$

• Define

• : product in $O^\uparrow(1,3)$

p : Rep of $O^\uparrow(1,3)$ acting on $\mathbb{R}^{1,3}$

$$p: O(1,3) \rightarrow \text{End}(\mathbb{R}^{4,3}) \left\{ \begin{array}{l} t: \text{in } \mathbb{R}^{3,3} \end{array} \right.$$

* Semi-direct group product: $\ltimes$

$$\Lambda_i \in O^\uparrow$$

$$a_i \in \mathbb{R}^{1,3}$$

$$(\Lambda_1, a_1) \ltimes (\Lambda_2, a_2)$$

$$(\Lambda_2, a_2) \circ (\Lambda_1, a_1) = (\Lambda_2 \Lambda_1, \rho(\Lambda_2) a_1 + a_2)$$

Representations of Poincare:

* Recall reps are homomorphisms:

$$U: [] \rightarrow \text{End}(\mathcal{H}) \quad * : \text{mult in } \text{End}(\mathcal{H})$$

$$U(g_1 * g_2) = U(g_1) * U(g_2)$$

$$U(\Lambda_2, a_2) \circ U(\Lambda_1, a_1) = U(\Lambda_2 \Lambda_1, \rho(\Lambda_2) a_1 + a_2)$$

• weaker: ρ : group homo for general $A \times B$.

* Connected Lie group: $T(\theta)$

• Have a finite set of real continuous parameters θ^a

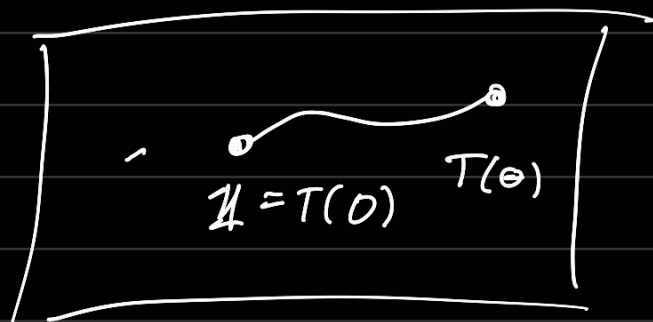
• Each element of the group connected to the $\mathbb{I}$ continuously

$$T(\bar{\theta}) T(\theta) = T(f(\bar{\theta}, \theta))$$

$$f^a(\bar{\theta}, 0)$$

$$T(\theta = 0) = \mathbb{I}$$

$$T(0) = \mathbb{I}$$



Thus

$$T(\bar{\theta}) = T(f(\bar{\theta}, 0))$$

$$\boxed{\begin{aligned} f(\bar{\theta}, 0) &= \bar{\theta} \\ f(0, \theta) &= \theta \end{aligned}}$$

$$* T(\theta) \in T_g G$$

$$U(T(\theta)) : \text{End}(T_g G)$$

$$U(T(\theta)) \approx 1 + \tilde{\theta}^\alpha t_\alpha + \frac{1}{2} \tilde{\theta}^\beta \tilde{\theta}^\alpha t_{\beta\alpha} + \dots$$

$$t_\alpha = \frac{\partial U}{\partial \theta^\alpha}$$

$$t_{\alpha\beta} = \frac{\partial U}{\partial \theta^\alpha \partial \theta^\beta}$$

$$t_{\alpha\beta} = t_{\beta\alpha}$$

$$\bullet U^\dagger = U$$

$$U^\dagger = 1$$

$$+ \dots$$

$$\bullet \theta^\alpha \text{ is an element } \in \mathbb{C}^n$$

$$\tilde{\theta}^\alpha \dagger t_\alpha^\dagger = \tilde{\theta}^\alpha t_\alpha$$

$$\bullet U(T(\theta)) \text{ form a rep. of the group:}$$

$$U(T(\bar{\theta})) U(T(\theta)) = U(T(f(\bar{\theta}, \theta)))$$

$$\begin{aligned} * f^a(\bar{\theta}, \theta) &= \sum_{n_1} \frac{1}{n_1!} \frac{\partial^{n_1} f^a(\bar{\theta}, 0)}{\partial \theta^{\beta n_1}} (\theta^\beta)^{n_1} \\ &= \sum_{n_1} \sum_n \frac{1}{n_1! n_2!} \partial_{\theta^\beta}^{n_1} \partial_{\bar{\theta}^\gamma}^{n_2} f^a(0, 0) (\bar{\theta}^\gamma)^{n_2} \theta^{\beta n_1} \end{aligned}$$

$$\left[\begin{array}{l} n_1 = 0 \\ n_2 = 0 \end{array} \right] \rightarrow \sum_{n_2} \frac{1}{n_2!} f$$

$$0 + \sum_{n_2} \partial_{\bar{\theta}^\gamma}^{n_2} f^a(0, 0) \bar{\theta}^{\gamma n_2}$$

$$\underline{n_2 = 1: \theta \text{ } n_1 = 0!} \quad \bar{\theta}^\gamma \partial_{\bar{\theta}^\gamma} f^a$$

$$f^a(\bar{\theta}, \theta) = f(\bar{\theta}, 0) + \theta^\beta \frac{\partial}{\partial \theta^\beta}$$

$$f^a(\bar{\theta}, 0) = f^a$$

$$\partial_{\theta^\beta}$$

$$f(\bar{\theta}, 0) = 0$$

then $f(\bar{\theta}, 0) = f(0, 0) + \bar{\theta}^\gamma \frac{\partial f}{\partial \bar{\theta}^\gamma} + \mathcal{O}(\bar{\theta}^2)$

(i) $f^a(0, 0) = 0$

(ii) $\frac{\partial f^a(0, 0)}{\partial \bar{\theta}^\gamma} = \delta_\gamma^\alpha$

likewise, $f(0, \theta) = f^a(0, 0) + \theta^B \frac{\partial f^a}{\partial \theta^B}$

$$\frac{\partial f^a(0, 0)}{\partial \theta^B} = \delta_B^\alpha$$

* Thus,

$$f(\bar{\theta}, \theta) = f^a(\bar{\theta}, \theta) + \theta^B \frac{\partial f^a}{\partial \theta^B}$$

$$= \bar{\theta}^\alpha + \theta^B \left(\frac{\partial}{\partial \theta^B} \left(f^a(0, 0) + \bar{\theta}^\gamma \frac{\partial f^a}{\partial \bar{\theta}^\gamma}(0, 0) \right) \right)$$

$$= \bar{\theta}^\alpha + \theta^B \left(\delta_B^\alpha + \bar{\theta}^\gamma \frac{\partial^2 f^a}{\partial \theta^B \partial \bar{\theta}^\gamma} \right)$$

$$= \bar{\theta}^\alpha + \theta^\alpha + \theta^B \bar{\theta}^\gamma \frac{\partial^2 f^a}{\partial \theta^B \partial \bar{\theta}^\gamma}$$

let $f_{\beta\gamma}^\alpha = \frac{\partial^2 f}{\partial \theta^\beta \partial \bar{\theta}^\gamma} \in \mathbb{R}$

$$f(\bar{\theta}, \theta) = \bar{\theta}^\alpha + \theta^\alpha + \theta^B \bar{\theta}^\gamma f_{\beta\gamma}^\alpha$$

$$U(T(\bar{\theta}))U(T(\theta)) = U(T(f(\bar{\theta}, \theta)))$$

$$(1 + i\bar{\theta}^a t_a + \frac{1}{2} \bar{\theta}^b \bar{\theta}^c t_{bc}) (1 + i\theta^r t_r + \frac{1}{2} \theta^\delta \theta^\epsilon t_{\delta\epsilon} + \dots)$$

$$= 1 + (\bar{\theta}^a + \theta^a + f_{\beta\gamma}^\alpha \theta^B \bar{\theta}^\gamma + \dots) t_a$$

$$+ (\bar{\theta}^a + \theta^a + f) (\theta^B + \bar{\theta}^B + f^B) t_{aB} + \dots$$

$$t_{bc} = -t_{cb} - i f_{bc}^\alpha t_a$$

t_{bc} must be symmetric (it's a second derivative)

$$t_{bc} = -t_b t_c - i f_{bc}^a t_a$$

$$t_{cb} = -t_c t_b - i f_{cb}^a t_a$$

$$t_{bc} - t_{cb} = 0$$

$$-t_b t_c - i f_{bc}^a t_a + t_c t_b + i f_{cb}^a t_a = 0$$

$$t_b t_c - t_c t_b = i (f_{cb}^a - f_{bc}^a) t_a$$

$$[t_b, t_c] = i C_{bc}^a t_a$$

• Structure constants

$$C_{bc}^a \equiv -f_{bc}^a + f_{cb}^a$$

• Abelian group: (trivial topology)

$$f^a(\bar{\theta}, \theta) = \bar{\theta}^a + \theta^a, \quad f = 0$$

and so

$$[t_a, t_b] = 0$$

*

$$U(T(\bar{\theta})) U(T(\theta)) = U(T(\theta + \bar{\theta}))$$

$$\bar{\theta} = \theta$$

$$U(T(\theta))^2 = U(T(2\theta))$$

$$U(T(\theta))^N = U(T(N\theta))$$

$$\tilde{\theta} \equiv \theta N, \quad \theta = \frac{\tilde{\theta}}{N}$$

$$U\left(T\left(\frac{\tilde{\theta}}{N}\right)\right)^N = U(T(\tilde{\theta}))$$

$$U = (A^N) \rightarrow A^N = A^N$$

$$U\left(T\left(\frac{\theta}{N}\right)\right) = \mathbb{1} + i \frac{\theta}{N} t_a$$

$$\lim_{N \rightarrow \infty} \left(\mathbb{1} + i \frac{\theta^\alpha t_\alpha}{N} \right)^N = U(T(\theta))$$

$$e^{i\theta^\alpha t_\alpha} = U(T(\theta))$$

$$* \quad \eta_{\mu\nu} dx^\mu dx^\nu = \eta_{\mu'\nu'} dx'^{\mu'} dx'^{\nu'}$$

$$\eta_{\mu\nu} = \frac{\partial x'^{\lambda}}{\partial x^\mu} \frac{\partial x'^{\sigma}}{\partial x^\nu} \eta_{\lambda\sigma}$$

$$\bullet \quad \left| \frac{d\vec{x}}{dt} \right| = 1 \Leftrightarrow d\vec{x}^2 - dt^2 = 0$$

$$\bullet \quad \text{Poincare':} \quad x'^{\mu} = \underbrace{x^\nu \Lambda_\nu^\mu}_{\mathcal{O}^\uparrow(1,3)} + \underbrace{a^\mu}_{\mathbb{R}^{1,3}}$$

$$\eta_{\mu\nu} x'^{\mu} x'^{\nu} = \eta_{\mu\nu} x^\mu x^\nu$$

$$\eta_{\mu\nu} \left(\Lambda_\sigma^\mu x^\sigma + a^\mu \right) \left(\Lambda_\lambda^\nu x^\lambda + a^\nu \right) = \eta_{\mu\nu} x^\mu x^\nu$$

$$\eta_{\mu\nu} \Lambda_\sigma^\mu x^\sigma \Lambda_\lambda^\nu x^\lambda + \eta_{\mu\nu} a^\nu \Lambda_\sigma^\mu x^\sigma + \eta_{\mu\nu} a^\mu \Lambda_\lambda^\nu x^\lambda + \eta_{\mu\nu} a^\mu a^\nu = \eta_{\mu\nu} x^\mu x^\nu$$

$$\eta_{\mu\nu} \Lambda_\sigma^\mu \Lambda_\lambda^\nu x^\sigma x^\lambda + 2 \eta_{\mu\nu} a^\mu \Lambda_\lambda^\nu x^\lambda + \eta_{\mu\nu} a^\mu a^\nu = \eta_{\mu\nu} x^\mu x^\nu$$

• Always possible to translate to a position s.t. $a^\mu = 0$

$$\eta_{\mu\nu} \Lambda_\sigma^\mu \Lambda_\lambda^\nu x^\sigma x^\lambda = \eta_{\sigma\lambda} x^\sigma x^\lambda$$

$$\therefore \boxed{\eta_{\mu\nu} \Lambda_\sigma^\mu \Lambda_\lambda^\nu = \eta_{\sigma\lambda}}$$

$$\Lambda_\tau^\nu \Lambda_\tau^\mu \eta_{\sigma\tau} = \eta^{\nu\mu}$$

*

$$T(\bar{\Lambda}, \bar{a}) T(\Lambda, a) = T(\bar{\Lambda} \Lambda, \bar{\Lambda} a + \bar{a})$$

$$\Rightarrow \eta \Lambda^2 = \eta$$

$$\det(\Lambda)^2 = 1 \Rightarrow \underline{\det(\Lambda) = \pm 1}$$

•

$$\underline{(\Lambda^{-1})^\mu_\sigma = \Lambda_\sigma^\mu = \eta_{\sigma\lambda} \Lambda^\lambda_\alpha \eta^{\mu\alpha}}$$

$$T(\Lambda, a) T(\alpha, \beta) = T(\Lambda\alpha, \alpha a + \beta)$$

$$T(\Lambda\alpha, \alpha a + \beta)$$

$$\bullet \Lambda\alpha = \mathbb{1}$$

$$\alpha = \Lambda^{-1}$$

$$\bullet \alpha a + \beta = 0$$

$$\beta = -\alpha a = -\Lambda^{-1} a$$

Thus

$$\boxed{(T(\Lambda, a))^{-1} = T(\Lambda^{-1}, -\Lambda^{-1} a)}$$

$$\mathbb{1} = T(\mathbb{1}, 0)$$

• $T(\Lambda, a)$: group element

$U(\Lambda, a)$: rep. of group

$$U(\Lambda, a) = \rho(T(\Lambda, a)) \in \text{End}(\mathcal{H})$$

$$\psi \longmapsto U(\Lambda, a) \psi$$

$$U(\bar{\Lambda}, \bar{a}) U(\Lambda, a) = U(\bar{\Lambda} \Lambda, \bar{\Lambda} a + \bar{a})$$

$$ISO^\uparrow(1,3) \cong \{T(\Lambda, a)\}$$

$$a=0: \cong O(1,3)^+$$

$$* \det(\Lambda) = 1$$

$$\eta_{\mu\nu} \Lambda^\mu_\sigma \Lambda^\nu_\tau = \eta_{\sigma\tau}$$

$$-\Lambda^0_\sigma \Lambda^0_\tau + \Lambda^i_\sigma \Lambda^i_\tau = \eta_{\sigma\tau}$$

$$(i) -(\Lambda^0_0)^2 + \Lambda^i_0 \Lambda^i_0 = -1$$

$$(ii) \eta^{\mu\nu} \Lambda^\mu_\sigma \Lambda^\nu_\tau = \eta^{\sigma\tau}$$

$$-\Lambda^0_\sigma \Lambda^0_\tau + \Lambda^i_\sigma \Lambda^i_\tau = -1$$

Thus

$$(\Lambda^0_0)^2 = 1 + \Lambda^i_0 \Lambda^i_0 = 1 + \Lambda^i_0 \Lambda^i_0$$

$$\det(\Lambda) = \epsilon_{\alpha_0 \dots \alpha_n} \Lambda^{\alpha_0}_0 \Lambda^{\alpha_1}_1 \dots \Lambda^{\alpha_n}_n = 1$$

$$(\Lambda^0_0) = \pm \sqrt{1 + \Lambda^i_0 \Lambda^i_0}$$

• Note $(\Lambda^i_0)^2$ is always positive so

$$\underline{(\Lambda^0_0) \geq 1} \quad \text{or} \quad \underline{(\Lambda^0_0) \leq -1}$$

↳ Form a subgroup

• $(\Lambda) \in (\bar{\Lambda})$:

$$(\bar{\Lambda} \Lambda)^\mu_\nu = \Lambda^\mu_\sigma \Lambda^\sigma_\nu$$

⇒

$$(\bar{\Lambda} \Lambda)^0_0 = \bar{\Lambda}^0_0 \Lambda^0_0 + \bar{\Lambda}^1_0 \Lambda^1_0 + \dots + \bar{\Lambda}^3_0 \Lambda^3_0$$

• Define the vector $V^i \equiv (\Lambda^1_0, \Lambda^2_0, \Lambda^3_0)$

$$V \cdot V = \|V\|^2 = \Lambda^i_0 \Lambda^i_0 = (\Lambda^0_0)^2 - 1$$

$$\|V\| = \sqrt{(\Lambda^0_0)^2 - 1}$$

• Similarly for $u \equiv (\Lambda_1^0, \dots, \Lambda_3^0)$
 $\|u\| = \sqrt{(\Lambda_0^0)^2 - 1}$

Cauchy-Schwarz:

$$|\langle u, v \rangle|^2 \leq \langle u, u \rangle \langle v, v \rangle$$

$$|\langle u, v \rangle|^2 \leq \|u\|^2 \|v\|^2$$

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

$$|\bar{\Lambda}_0^0 \Lambda_0^0 + \dots + \bar{\Lambda}_0^0 \Lambda_0^0| \leq \sqrt{(\Lambda_0^0)^2 - 1} \sqrt{(\bar{\Lambda}_0^0)^2 - 1}$$

$$(\bar{\Lambda} \Lambda)_0^0 \geq \bar{\Lambda}_0^0 \Lambda_0^0 - \sqrt{\Lambda_0^{0^2} - 1} \sqrt{\bar{\Lambda}_0^0 - 1} \stackrel{?}{\geq} 1$$

$$\underbrace{(\Lambda_0^0)^2 \geq 1}_{\gamma} \quad \underbrace{\Lambda_0^{0^2} - 1 \geq 0}_{\alpha\beta \geq 0} \quad ?$$

$$(\bar{\Lambda}_0^0)^2 (\Lambda_0^0)^2 - \Lambda_0^{0^2} - \bar{\Lambda}_0^{0^2} + 1$$

$$|A_0 + \langle u, v \rangle| \leq |A| + |\langle u, v \rangle|$$

$$\leq |A| + \|u\| \|v\|$$

$$|A - \langle u, v \rangle| \leq |A_0 + \langle u, v \rangle| \leq |A| + \|u\| \|v\|$$

not sure but

$$(\bar{\Lambda} \Lambda)_0^0 \geq 1$$

Proper orthochronous Lorentz group:

$$\uparrow$$

$$ISO(1, 3)^+ \equiv \{ \Lambda \in O(1, 3) \mid \det(\Lambda) = 1, \Lambda_0^0 \geq 1 \}$$

$$\Downarrow$$

$$\mathbb{I}$$

$\mathbb{P}$: Space of Inversions: (Parity)

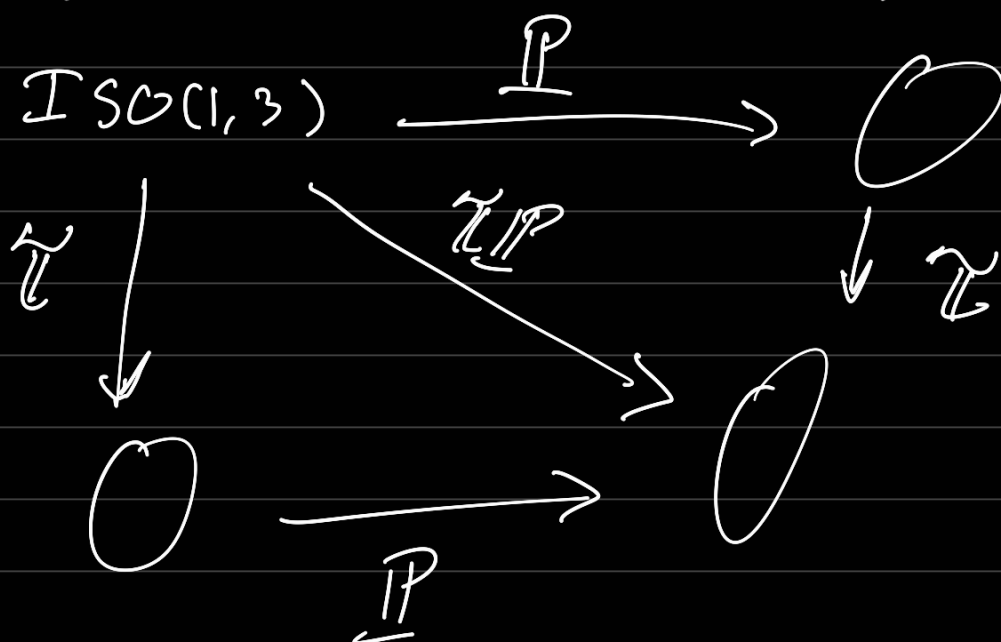
$$\mathbb{P}_0^0 = 1 \quad \mathbb{P}_1^2 = \mathbb{P}_2^2 = \mathbb{P}_3^3 = -1$$

$$\boxed{P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}} \quad - \eta_{\mu\nu}$$

$\mathcal{T}$: Time reversal: $\mathcal{T}_0^0 = -1, \mathcal{T}_i^i = 1$

$$\boxed{\mathcal{T} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}} \quad \sim \eta_{\mu\nu}$$

(Lorentz) $\longrightarrow$ (Prop. orthogonal)
 $O(1,3)$ $ISO(1,3) = \oplus \{P, \mathcal{T}\}$



* 4 disconnected components

Q: What if there's a deeper meaning to the signature of our metrics. It seems

$$\left\{ \begin{array}{l} \eta_{ret} \sim \mathcal{T} \text{ (time reversal)} \\ \eta_{ps} \sim P \text{ (parity)} \end{array} \right\}$$

$$T(1,0) = \emptyset$$

or $\Lambda_{\mu\nu}^M = \delta_{\mu\nu}^M$ and $g^{\mu\nu} = 0$

• Want to look at

$$\Lambda_{\mu\nu}^M = \delta_{\mu\nu}^M + \omega_{\mu\nu}^M, \quad g^{\mu\nu} = \epsilon^{\mu\nu}$$

$$|\omega|, |\epsilon| \ll 1$$



$$\eta_{\sigma\tau} = \eta_{\mu\nu} (\delta_{\sigma}^{\mu} + \omega_{\sigma}^{\mu}) (\delta_{\tau}^{\nu} + \omega_{\tau}^{\nu})$$

$$\eta_{\sigma\tau} = \eta_{\mu\nu} (\delta_{\sigma}^{\mu} \delta_{\tau}^{\nu} + \omega_{\sigma}^{\mu} \delta_{\tau}^{\nu} + \delta_{\sigma}^{\mu} \omega_{\tau}^{\nu} + \cancel{\omega_{\sigma}^{\mu} \omega_{\tau}^{\nu}})$$

$$= \eta_{\sigma\tau} + \omega_{\tau\sigma} + \omega_{\sigma\tau} + \cancel{\omega^2}$$

$$\boxed{\omega_{\tau\sigma} = -\omega_{\sigma\tau}}$$

$$\bullet \omega \in \Lambda^2(G) : \quad \frac{4 \cdot 3}{2} = \underline{6} \quad \binom{4}{2} \quad \text{D.O.F}$$

$$(\omega_{\mu\nu} \oplus a_{\mu})$$

$$\underline{6} \oplus \underline{4} = 10 \text{ parameters}$$

$$ISO(1,3)^+ \rightarrow \dim(\mathfrak{g}) = \underline{10}$$

$$\ast U(1 + \omega, \epsilon) = 1 + \frac{1}{2} i \omega_{\rho\sigma} J^{\rho\sigma} = i \epsilon_{\rho} P^{\rho} + \dots$$

$$J^{\rho\sigma} = -J^{\sigma\rho}$$

$$\ast U(1, a) : \text{translations}$$

$$T(1, \bar{a}) T(1, a) = T(1, \bar{a} + a)$$

$$U(1, \bar{a}) U(1, a) = U(1, \bar{a} + a)$$

$$a = \bar{a}$$

$$(U(1, a))^2 = U(1, 2a)$$

$$(U(1, a))^N = U(1, Na)$$

$$\alpha \equiv Na$$

$$\left(U(1, \frac{\alpha}{N}) \right)^N = U(1, \alpha)$$

$$U(1, \frac{\alpha}{N})^N = \left(1 - \frac{i \alpha_{\mu\nu}}{N} P^{\mu\nu} \right)^N$$

$$\lim_{N \rightarrow \infty} \left(1 - i \frac{a_\mu p^\mu}{N} \right) = U(1, \alpha)$$

$$U(1, \alpha) = e^{-i \hat{p}^\mu a_\mu}$$

* Rotations: $R_\theta \equiv R[\theta]$ |2|

$$U(R_\theta, 0) = \exp(i \mathbf{J} \cdot \boldsymbol{\theta})$$

One Particle States:

• $\psi(p, \sigma) \rightarrow \langle p, \sigma | \psi \rangle$ all other D.O.F (spin, helicity, ...)

$$\hat{p}^\mu \psi_{p, \sigma} = p^\mu \psi_{p, \sigma}$$

$$U(1, a) \psi_{p, \sigma} = e^{-i p^\mu a_\mu} \psi_{p, \sigma}$$

Must now understand how $U(\Lambda, 0)$ (Lorentz) acts on this state

$$U(\Lambda) \equiv U(\Lambda, 0)$$

$$\hat{p}^\mu U(\Lambda) \psi_{p, \sigma} = U(\Lambda) (U^{-1}(\Lambda) \hat{p}^\mu U(\Lambda)) \psi_{p, \sigma}$$

but

$$\begin{cases} U(\Lambda) \hat{p}^\mu U^{-1}(\Lambda) = \Lambda^\mu_\rho \hat{p}^\rho \\ U^{-1} \hat{p}^\mu U = (\Lambda^{-1})^\mu_\rho \hat{p}^\rho \\ \quad \quad \quad = \Lambda^\rho_\mu \hat{p}^\rho \end{cases}$$

$$U(\Lambda) (\Lambda^\mu_\rho \hat{p}^\rho) \psi_{p, \sigma} = \Lambda^\mu_\rho p^\rho U(\Lambda) \psi_{p, \sigma}$$

* This tells us that

$$U(\Lambda) \psi_{p, \sigma}$$

has momenta

$$\Lambda^\mu_\rho p^\rho$$

so

$$U(\Lambda) \psi_{p,\sigma} \sim \psi_{\Lambda p, \sigma'}$$

or

$$U(\Lambda) \psi_{p,\sigma} = \sum_{\sigma'} C_{\sigma'\sigma}(\Lambda, p) \psi_{\Lambda p, \sigma'}$$

→ Can look at irreducible reps of $ISO(1,3)$ to understand how $C_{\sigma'\sigma}$ looks.

• only function of p^μ which is invariant of Λ^μ_ν is

$$p^2 = p^\mu p^\nu \eta_{\mu\nu} = -m^2$$

$$p^2 \leq 0:$$

$$f(p) \stackrel{!}{=} p^2$$

• For each mass p^2 and p^0 :
choose a standard momentum, k^μ

$$p^\mu = L^\mu_\nu(p) k^\nu$$

Define: the states $\psi_{p,\sigma}$ of momentum p by:

$$\psi_{p,\sigma} \equiv N(p) U(L(p)) \psi_{k,\sigma}$$

• $N(p)$: numerical normalisation factor

$$\begin{aligned} U(\Lambda) \psi_{p,\sigma} &= N(p) U(\Lambda) U(L(p)) \psi_{k,\sigma} \\ &= N(p) U(\Lambda L(p)) \psi_{k,\sigma} \end{aligned}$$

insert

$$1 = U(L(\Lambda p)) U^{-1}(L(\Lambda p))$$

$$U(\Lambda) \psi_{p,\sigma} = N(p) U(L(\Lambda p)) U(L^{-1}(\Lambda p) \Lambda L(p)) \psi_{k,\sigma}$$

$$U(L(p)) k = 0$$

$$\Delta(L(p)K) = \Delta p$$

$$L^{-1}(\Delta p)(\Delta p) = K$$

So

$$L^{-1}(\Delta p) \Delta L(p) = \mathbb{1} \in \text{L.G. of } ISO(1;3)$$

s.t

$$W_{\nu}^{\mu} K^{\nu} = K^{\mu} \quad \Big| \quad \begin{matrix} U \\ W \end{matrix}$$

$W_{\nu}^{\mu} \in \text{Little group}$

$$U(W) \psi_{k,\sigma} = \sum_{\sigma'} D_{\sigma'\sigma}(W) \psi_{k,\sigma'}$$

$D_{\sigma'\sigma}(W)$: Furnish a rep. of the little group

$$\begin{aligned} \sum_{\sigma'} D_{\sigma'\sigma}(\bar{W}W) \psi_{k,\sigma'} &= U(\bar{W}W) \psi_{k,\sigma} \\ &= U(\bar{W}) U(W) \psi_{k,\sigma} \\ &= U(\bar{W}) \sum_{\sigma''} D_{\sigma''\sigma}(W) \psi_{k,\sigma''} \\ &= \sum_{\sigma''} D_{\sigma'\sigma''}(\bar{W}) D_{\sigma''\sigma}(W) \psi_{k,\sigma''} \end{aligned}$$

$$D_{\sigma'\sigma}(\bar{W}W) = \sum_{\sigma''} D_{\sigma'\sigma''}(\bar{W}) D_{\sigma''\sigma}(W)$$

$$* W(\Delta, p) \equiv L^{-1}(\Delta p) \Delta L(p)$$

$$\begin{aligned} U(\Delta) \psi_{p,\sigma} &= N(p) U(L(\Delta p)) U(W(\Delta, p)) \psi_{k,\sigma} \\ &= N(p) U(L(\Delta p)) \sum_{\sigma'} D_{\sigma'\sigma}(W) \psi_{k,\sigma'} \\ &= N(p) \sum_{\sigma'} D_{\sigma'\sigma}(W) \underline{U(L(\Delta p))} \psi_{k,\sigma'} \end{aligned}$$

Since

$$\psi_{p,\sigma} = N(p) U(L(p)) \psi_{k,\sigma}$$

then

$$\psi_{\Delta p,\sigma} = N(\Delta p) U(L(\Delta p)) \psi_{k,\sigma}$$

$$U(L(\Delta p)) \psi_{k,\sigma} = \frac{\psi_{\Delta p,\sigma'}}{N(\Delta p)}$$

$$U(\Delta) \psi_{p,\sigma} = \left(\frac{N(p)}{N(\Delta p)} \right) \sum_{\sigma'} D_{\sigma\sigma'}(W(\Delta, p)) \psi_{\Delta p,\sigma'}$$

* Acting on $\psi_{p,\sigma}$ with $U(\Delta)$ gives an expansion of $\psi_{\Delta p,\sigma}$ w/ coeff. of a rep of the little group of the particle.

$$= \sum_{\sigma'} C_{\sigma\sigma'}(\Delta) \psi_{\Delta p,\sigma'}$$

* Finding $C_{\sigma\sigma'}$ has been reduced to finding (irreps) of the little group
Method of induced reps:

	K^M	Little Group
$p^2 = -m^2 < 0, p^0 > 0$	$(M, 0, 0, 0)$	$SO(3)$
$p^2 = -m^2 < 0, p^0 < 0$	$(-M, 0, 0, 0)$	$SO(3)$
$p^2 = 0, p^0 > 0$	$(k, 0, 0, k)$	$ISO(2)$
$p^2 = 0, p^0 < 0$	$(k, 0, 0, -k)$	$ISO(2)$
$p^2 = N^2 > 0$	$(0, 0, N, 0)$	$SO(?, 1)$
$p^M = 0$	$(0, 0, 0, 0)$	$SO(3, 1)$

$ISO(2)$: Euclidean group: $(\mathbb{R} \oplus \mathbb{R}^2)$

* $(a), (c), (f)$ are the only physical states

(f) $p^M = 0$: Vacuum is trivially invariant under $U(\Delta)$.

(a) : Massive reps

(c) : Massless

(c): massless reps

* Normalization of states:

$$\langle \psi_{k'\sigma'}, \psi_{k\sigma} \rangle = \delta^3(k' - k) \delta_{\sigma\sigma'}$$

$$U(W) \psi_{k\sigma} = \sum_{\sigma'} D_{\sigma'\sigma}(W) \psi_{k\sigma'}$$

$$\langle U(W) \psi_{k'\sigma'}, U(W) \psi_{k\sigma} \rangle =$$

$$= \langle \sum_{\sigma''} D_{\sigma''\sigma'}(W) \psi_{k'\sigma''}, \sum_{\tilde{\sigma}} D_{\tilde{\sigma}\sigma}(W) \psi_{k\tilde{\sigma}} \rangle$$

$$= \sum_{\sigma''} \sum_{\tilde{\sigma}} D_{\sigma''\sigma'}^\dagger(W) D_{\tilde{\sigma}\sigma}(W) \langle \psi_{k'\sigma''}, \psi_{k\tilde{\sigma}} \rangle$$

$$= \sum_{\sigma''} D_{\sigma''\sigma'}^\dagger(W) D_{\sigma''\sigma}(W) \delta^3(k' - k)$$

$$= \sum_{\sigma''} D_{\sigma''\sigma'}(W^\dagger) D_{\sigma''\sigma}(W) \delta^3(k' - k)$$

$$= D_{\sigma'\sigma}(W^\dagger W) \delta^3(k' - k)$$

* In order for

$$D_{\sigma'\sigma}(W^\dagger W) = \delta_{\sigma\sigma'}$$

Require

$$W^\dagger W = \mathbb{1}$$

Thus W is unitary and so too is

$$D^\dagger(W) D(W) = \mathbb{1}$$

Now, to find $N(p)$:

$$\begin{aligned} \langle \psi_{p'\sigma'}, \psi_{p\sigma} \rangle &= \langle \psi_{p'\sigma'}, N(p) U(L(p)) \psi_{k,\sigma} \rangle \\ &= N(p) \langle U^{-1}(L(p)) \psi_{p'\sigma'}, \psi_{k\sigma} \rangle \end{aligned}$$

*

$$U(L^{-1}(p)) \psi_{p'\sigma'}$$

$$= \frac{N(p')}{N(L^{-1}(p)p)} \sum_{\sigma''} D_{\sigma''\sigma'}(W(L^{-1}(p), p')) \psi_{k'\sigma''}$$

Note $k = L^{-1}(p)p'$
 ≤ 0

$$\psi_{p\sigma} = N(p) U(L(p)) \psi_{k\sigma}$$

$$k' = L^{-1}(p)p'$$

$$\psi_{k'\sigma} = N(k') U(L^{-1}(p)p') \psi_{k\sigma}$$

$$\psi_{p'\sigma'} = N(p') U(L(p)) \psi_{k'\sigma'}$$

$$U(L^{-1}(p)) \psi_{p'\sigma'} = \frac{N(p')}{N(L^{-1}(p)p')} \sum_{\sigma''} D_{\sigma''\sigma'}(W(L^{-1}(p), p')) \times \psi_{k'\sigma''}$$

$$N(k') = \mathbb{I} ???$$

$$\hookrightarrow = N(p) \left(\frac{N(p')}{N(k')} \sum_{\sigma''} D_{\sigma''\sigma'}(W(L^{-1}(p), p')) \psi_{k'\sigma''}, \psi_{k\sigma} \right)$$

$$= \frac{N(p) N^*(p')}{N(k')} \sum_{\sigma''} D_{\sigma''\sigma'}^*(W(L^{-1}(p), p')) (\psi_{k'\sigma''}, \psi_{k\sigma})$$

$$= \frac{N(p) N^*(p')}{N(k')} \sum_{\sigma''} D_{\sigma''\sigma'}^*(W(L^{-1}(p), p')) \delta^3(k' - k) \delta_{\sigma''\sigma}$$

$$= \frac{N(p) N^*(p')}{N(k')} D_{\sigma\sigma'}^*(W(L^{-1}(p), p')) \delta^3(k' - k)$$

• Not sure where $N(k')$ goes ...
 should be $N(p) N^*(p')$

$$k' = L^{-1}(p)p' \mid k = L^{-1}(p)p$$

• $p = p'$: $W(L^{-1}(p), p) = \mathbb{I}$

$$D_{\alpha\alpha'}(q) = \delta_{\alpha\alpha'}$$

$$(\psi_{p'\alpha'}, \psi_{p\alpha}) = |N(p)|^2 \delta_{\alpha\alpha'} \delta^3(k' - k)$$

* Relationship between $\delta^3(k' - k)$ & $\delta^3(p' - p)$
 $-p^2 = M^2 \geq 0, p^0 > 0$

• Lorentz invariant measure:

$$\int d^4p \delta(p^2 + M^2) \Theta(p^0) f(p)$$

where

$$\Theta(x) \equiv \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Since

$$\int_0^{\infty} dp^0 \quad \text{and} \quad \underbrace{\int_{-\infty}^0 dp^0}_{\text{zero}}$$

$$\text{thus, } \int_{-\infty}^{\infty} dp^0 \Theta(p^0)$$

Note

$$\frac{d\Theta}{dp^0} = \delta(p^0)$$

* Claim:

$$d^4k \delta(p^2 + M^2) \Theta(p^0)$$

is the Lorentz invariant measure

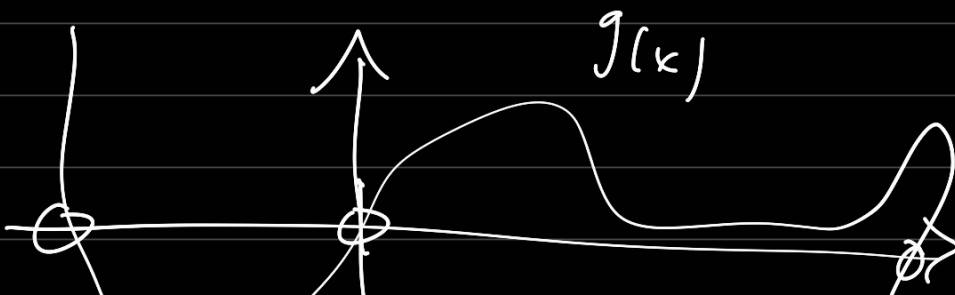
Take

$$\int f(x) \delta(x) dx = f(0)$$

$$\bullet \int f(x) \delta(g(x)) dx$$

$$z = g(x) \Rightarrow dz = g'(x) dx$$

$$\int f(x) \delta(g(x)) dx = \int f(z) \delta(z) \frac{dz}{g'(x)}$$



$$\int_{-\infty}^{\infty} f(x) \delta(g'(x_i)x) dx$$

$$f(g(x_i)) \quad (\text{near } x_i)$$

$$g(x) - g(x_i) \approx g'(x_i)(x - x_i)$$

$$g(x) \approx g'(x_i)(x - x_i)$$

$$\sum_i \int_{\epsilon_i}^{\bar{\epsilon}_i} f(y) \delta(y) \frac{dy}{|g'(x_i)|}$$

$$= \sum_i \frac{1}{|g'(x_i)|} \int_{\epsilon_i}^{\bar{\epsilon}_i} f(x) \delta(y) dy$$

$$= \sum_i \frac{f(x_i)}{|g'(x_i)|}$$

$$\underline{f=1:}$$

$$\delta(g(x)) = \sum_i \frac{\delta(x - x_i)}{|g'(x_i)|}$$

$$\int_{-\infty}^{\infty} \delta(g(x)) dx = \sum_i \int_{\epsilon_i}^{\epsilon_i + \Delta} \delta(g'(x_i)(x - x_i)) dx$$

$$y = (x - x_i)$$

$$\sum_i \int \delta(g'(x_i)y) dy$$

$$= \sum_i \int \delta(z) \frac{dz}{|g'(x_i)|}$$

$$x_i = y = x \leftrightarrow x_i$$

$$\rightarrow \sum_i \int_{x_i}^{x_i+\Delta} \frac{\delta(x-x_i)}{|g'(x_i)|} dx$$

$$\therefore \delta(g(x)) = \sum_i \frac{\delta(x-x_i)}{g'(x_i)}$$

$$\Rightarrow \delta(k^2 + m^2) = \delta(-k_0^2 + k_i^2 + m^2)$$

$$\int_{-\infty}^{\infty} d^4 p \delta(p^2 + m^2) \Theta(p^0) = \frac{1}{\left| \frac{\partial}{\partial p^0} (p^2 + m^2) \right|}$$

$$\left| \frac{\partial}{\partial p^0} (-p^0^2 + p_i^2 + m^2) \right| = |-2p^0|$$

$$\int_{-\infty}^{\infty} dp^0 \delta(p^2 + m^2) \Theta(p^0) = \frac{1}{2p^0}$$

* Cool, now,

$$\int d^4 p \delta(p^2 + m^2) \Theta(p^0) f(p)$$

$$= \int d^3 \vec{p} dp^0 \delta((p^0)^2 - \vec{p}^2 - m^2) \Theta(p^0) f(p)$$

$$= \int d^3 \vec{p} \frac{f(\vec{p}, \sqrt{\vec{p}^2 + m^2})}{2\sqrt{\vec{p}^2 + m^2}}$$

• on the mass-shell: $p^2 + m^2 = 0$ the invariant volume element is

$$\boxed{\frac{d^3 \vec{p}}{\sqrt{\vec{p}^2 + m^2}}}$$

$$F(\vec{p}) = \int F(\vec{p}') \delta^3(\vec{p}' - \vec{p}) d^3 \vec{p}'$$

$$= \int (F(\vec{p}') \sqrt{\vec{p}'^2 + m^2}) \delta^3(\vec{p}' - \vec{p}) \left(\frac{d^3 \vec{p}'}{\sqrt{\vec{p}'^2 + m^2}} \right)$$

- So, in order that the above be fully invariant

$$\sqrt{\vec{p}^2 + m^2} \delta^3(\vec{p}' - \vec{p}) = p^0 \delta^3(\vec{p}' - \vec{p})$$

(invariant Delta-fn)

- Measure would differ by $\det(L)$ but $\det(L) = 1$ so

$$k^0 \delta^3(\vec{k}' - \vec{k}) = p^0 \delta^3(\vec{p}' - \vec{p})$$

$$\begin{aligned} (\psi_{p'\sigma'}, \psi_{p\sigma}) &= |N(p)|^2 \delta_{\sigma'\sigma} \delta^3(\vec{k}' - \vec{k}) \\ &= |N(p)|^2 \delta_{\sigma'\sigma} \left(\frac{p^0}{k^0} \right) \delta^3(\vec{p}' - \vec{p}) \end{aligned}$$

* Normalize s.t.

$$|N(p)|^2 \left(\frac{p^0}{k^0} \right) = 1 \Rightarrow N(p) = \sqrt{\frac{k^0}{p^0}}$$

$$(\psi_{p'\sigma'}, \psi_{p\sigma}) = \delta_{\sigma'\sigma} \delta^3(\vec{p}' - \vec{p})$$

* Look at Massive / Massless reps